Image size (`imgsz`) compared with the baseline

Tuned by Ben Nichiporik · YOLOv8s gesture detector · imgsz 480 / 800 / 960 vs baseline 640 · November 2026

0.848
best team mAP50 (imgsz 800)
vs 0.845 baseline: +0.003

±0.017
largest change in team mAP50
(single-model noise ±0.06)

8 / 9 / 10 / 9
small hands found of 16
at imgsz 480 / 640 / 800 / 960

0.821
baseline F1 at conf 0.25,
best of all four runs

Result: changing the image size did not improve the baseline. Team-only mAP50 moved by at most 0.017, well inside the ±0.06 uncertainty of a single model on our 128 photos. imgsz 800 is best on paper, but the baseline (640) keeps the best F1 at the deployment threshold and the best speed/accuracy balance. A larger input did not help small hands, because the baseline's problem is unfamiliar *scenes*, not too few pixels.

1. Setup and hypothesis

`imgsz` is the size (longest side, in pixels) that every image is resized to for training and inference. The baseline evaluation showed that the model's main weakness on our own photos is **small hands**: only 56% of the 16 hands covering under 1% of the image were found. At imgsz 640 those hands are only about 25-62 px across (median hand 119 px).

Hypothesis. A larger `imgsz` (800, 960) keeps more pixels per hand and should raise recall on small hands, especially fists, at the cost of slower training and inference. A smaller one (480) shows how much accuracy we lose for speed, which matters for a real-time photo booth.

run	imgsz	epochs run	best epoch	train min
imgsz_480	480	50	41	5.3
baseline	640	50	41	8.5*
imgsz_800	800	50	38	17.0
imgsz_960	960	41	26	26.0

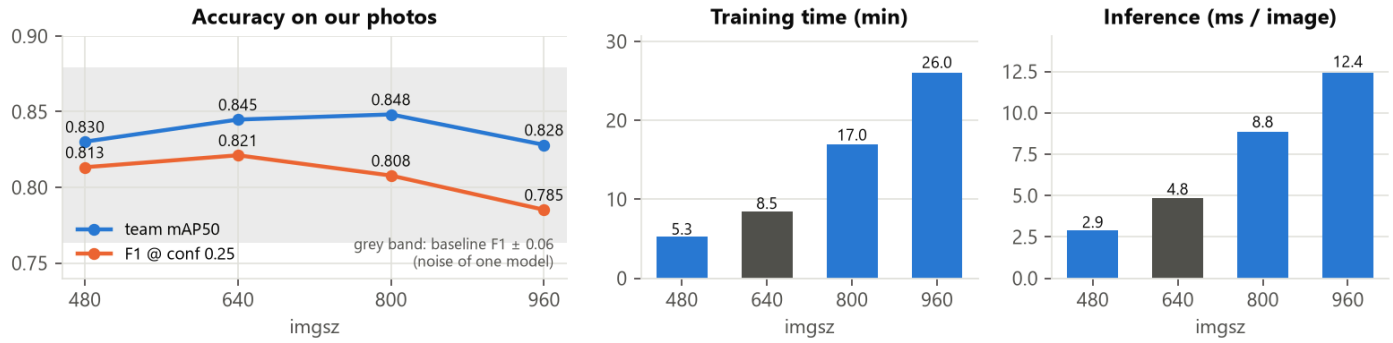
Only `imgsz` changes. Everything else is the baseline: YOLOv8s (COCO), SGD lr0 0.01, 50 epochs, patience 15, batch 16, seed 0, deterministic. *8.5 min on an idle GPU (15.3 min in the final re-run on a shared GPU, identical model).

Fair comparison. Same split, seed and evaluation code (`gesture_eval.py`). Each model is scored at the size it was trained at, and the headline is our own photos only: 128 images, 154 hands, at confidence ≥ 0.25 and IoU ≥ 0.5.

2. Results compared with the baseline

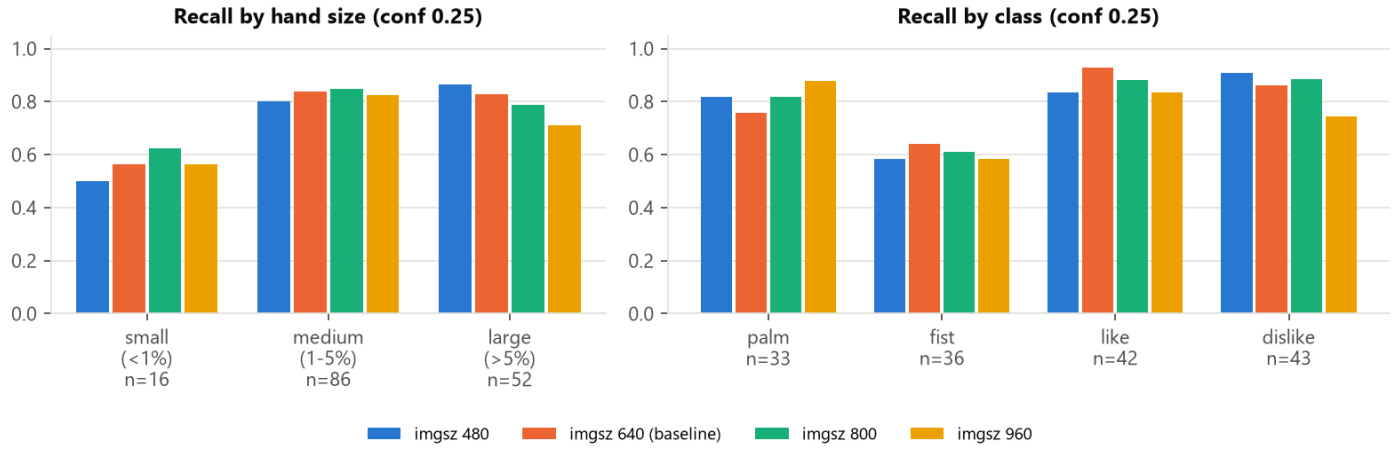
imgsz	val mAP50	team mAP50	team mAP50-95	fist mAP50	precision	recall	F1 @ 0.25	Δ team mAP50	Δ F1 @ 0.25	small hands	fixed / broken
480	0.995	0.830	0.426	0.618	0.855	0.780	0.813	−0.015	−0.008	8 / 16	9 / 11
640 (baseline)	0.990	0.845	0.418	0.625	0.894	0.775	0.821	—	—	9 / 16	—
800	0.994	0.848	0.429	0.642	0.894	0.765	0.808	+0.003	−0.013	10 / 16	9 / 9
960	0.995	0.828	0.426	0.594	0.880	0.751	0.785	−0.017	−0.036	9 / 16	12 / 19

mAP, precision and recall from Ultralytics on our photos (val = HaGRID validation). F1, small hands (<1% of the image) and fixed / broken (hands the baseline got wrong that this run gets right / the reverse) at conf 0.25, IoU 0.5.



Grey band: baseline F1 ± 0.06, the 95% bootstrap half-width of one model's F1 on our photos. Inference is per image at batch 16 on an RTX 5060 Ti and varies by 1-2 ms between runs.

- **No run beats the baseline by more than noise.** Validation is saturated (0.990-0.995) for every size, so it can't separate the runs either.
- **imgsz 800 is best on paper, but only just**: highest team mAP50 (0.848, +0.003), mAP50-95 (0.429) and fist mAP50 (0.642, +0.018). Its precision is the same as the baseline's and its recall is lower (0.765 vs 0.775), so F1 at the deployment threshold is slightly *worse* (0.808 vs 0.821). It fixed 9 hands and broke 9.
- **imgsz 960 is the worst for its cost.** It early-stopped at epoch 41 (best epoch 26), took 3× the baseline's training time, fixed 12 hands but broke 19, and large-hand recall fell from 0.83 to 0.71.
- **imgsz 480** is the fastest and only slightly worse (−0.015 mAP50), but it finds the fewest small hands (8 / 16).



Small hands barely moved. 8/16 at 480, 9/16 at 640, 10/16 at 800, 9/16 at 960. One extra hand at 800 is not a real improvement on a sample this small, so a larger `imgsz` did not fix the baseline's main weakness. Per class, fist stays the weakest class at every size (recall 0.58, 0.64, 0.61, 0.58).

3. Why a larger image size didn't help

Resolution was not the real problem. The training set already has plenty of small hands: 386 of the 1,280 HaGRID training hands (30%) cover under 1% of the image, and the median training hand (1.6% of the image) is *smaller* than ours (3.5%). The gap is **scene context**. HaGRID shows one person standing and facing a phone camera, mostly in a plain indoor room. Our missed hands are in crowds, outdoors, at odd angles and against busy backgrounds. A larger input gives the model more pixels, but it doesn't teach it what those scenes look like.

imgsz 960 may be under-trained. Its best checkpoint is from epoch 26, so it effectively trained for about half as long as the others. That probably explains part of its drop on large hands.

imgsz	baseline weights run at this size	model trained at this size
480	0.837	0.830
640 (baseline)	0.845	0.845
800	0.823	0.848
960	0.785	0.828

Control: team mAP50 of the baseline weights (trained at 640) run at each size, compared with a model trained at that size.

The control fits this explanation. Running the 640-trained baseline at a larger size *lowers* team mAP50 (0.845 → 0.823 at 800 → 0.785 at 960). Our hands are already larger on average than HaGRID's, and upscaling pushes them further from the sizes seen in training. Training at the larger size only wins back what the mismatch cost.

Label errors. A few of our test labels contain human error (loose boxes on Ben's `fist_380/381/384`, unlabelled raised hands in `fist_376`, `fist_382` and `palm_388`, an unclear gesture in `palm_376`, and the duplicate pair `dislike_388` = `dislike_393`). For the baseline they cause 3 of the 24 missed hands and 4 of the 18 background predictions. Every run is scored against the same labels, so the comparison between sizes is still fair, but absolute scores (fist especially) are slightly pessimistic.

4. Choice and next steps

1. **For the group comparison, my best value is imgsz 800** (best team mAP50, mAP50-95 and fist mAP50). The gain over the baseline is +0.003 mAP50, effectively a tie, and it costs about 2× the training time and 1.8× the inference time.
2. **For the photo booth, keep imgsz 640.** It has the best F1 at the deployment threshold (0.821) and the best balance of speed and accuracy.
3. **Fix the data, not the resolution.** More training photos with varied scenes (crowds, outdoors, several people) and augmentation that changes the scene (`mosaic`, `mixup`, `scale`, colour changes) target the real weakness. If another member's hyperparameter improves scale robustness (e.g. `scale` or `mosaic`), combining it with imgsz 800 is the combination worth testing.